PLANCKS Dublin 2024

Trinity College Dublin
25th May 2024

This exam contains 38 pages (including this cover page) and 10 exercises.
You are required to show your work on each problem on this exam. The following rules apply:

- This paper consists of **10 problems, each worth a total of 10 marks**. For questions with multiple parts, the subdivisions of marks are indicated.
- The exam will **start at 11:30 am**. Participants have a total working time of **four hours** to complete the paper.
- The use of non-medical hardware (including phones, tablets, etc.) and external sources (including textbooks, non-team members, the internet etc.) is not permitted. Scientific, non-programmable and non-graphing calculators, watches and medical equipment are allowed.
- You may use a dictionary: English to your native language.
- Please leave your phones and devices in the envelope at the start to be collected. They will be handed back to you at the end of the exam.
- When a problem is unclear, a participant can ask, via the invigilators, for a clarification. If the response is relevant to all teams, it will be provided to the other teams.
- Organise your work, in a reasonably neat and coherent way. Indicate at the top of each page your UIC, the question number, and the page number for that question.
- Place all problems in the envelope in order at the end of the exam. If your answer to a question has multiple pages, please staple the pages together for each question.
- If it is brought to the invigilators' attention that a team has been cheating, breaking the rules or misbehaving, they will be disqualified. Disqualification can happen post submission.
- In situations to which no rule applies, the organisation will rule on the matter following consultation with the UK & Ireland's PLANCKS Advisory Board, who serve as the jury for PLANCKS Dublin 2024.

PLANCKS Dublin 2024 Exam

Question 1: Moment of Inertia

Question 2: Quaternions

Question 3: Möbius Strip.... Time Travel?

Question 4: Icy Roof

Question 5: Floating Ring

Question 6: Nuclear Medicine Scan

Question 7: Dark Matter in the Galaxy

Question 8: Icosahedron of Resistors

Question 9: Molecular Zipper

Question 10: 4D Sun

In case you have forgotten for this **PLANCKS** exam: Planck's constant, $h = 1$. However, if you need it in actual experimentalists' units $h = 6.63 \times 10^{-34}$ Js.

1 Moment of Inertia

This question concerns the moment of inertia, I_{G}, of a uniform lamina and mass m, about an axis perpendicular to the plane of the lamina, passing through the mass centre, G.

Figure 1 shows ABC is an isosceles triangle with two sides length r and third side $2a$.

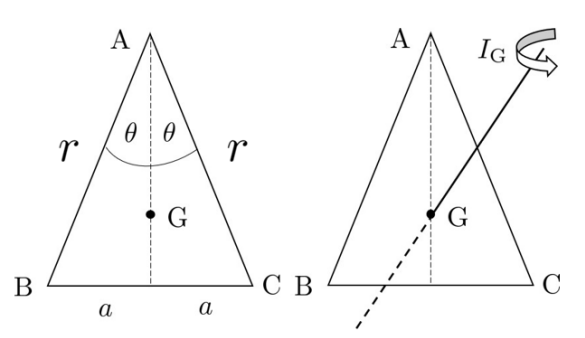


Figure 1: Isosceles triangle with semivertical angle θ and moment of inertia I_G.

(a) [6 marks] Show that the formula for the moment of inertia about the centre of mass is

$$I_{\mathrm{G}} = \frac{1}{18} mr^2 \left(2 - \cos 2\theta\right).$$

Consider now a regular $n-$gon lamina comprised of n isosceles triangles. The diameter of the $n-$gon will be $2r$ and its mass centre is denoted as G$'$. You may take the total mass of the $n-$gon lamina to be M.

For example, a $6-$gon lamina (i.e a hexagon) would be formed of six isosceles triangles, with their apexes touching. Each of these triangles has two sides of length r.

(b) [4 marks] Find an expression for the moment of inertia of a regular $n-$gon lamina about its centre of mass G$'$ and show that in the limit $n \to \infty$ the result for a circle is recovered i.e.

$$\frac{1}{2} Mr^2.$$

2 Quaternions

Everybody[a] knows how complex numbers work. Each complex number is made of a doublet of real numbers:

$$\begin{aligned} z_1 &= a_1 + ib_1 \\ z_2 &= a_2 + ib_2 \end{aligned}$$

where $i^2 = -1$. In order to be useful, one has to be able to add and multiply complex numbers:

$$\begin{aligned} z_1 + z_2 &= (a_1 + a_2) + i(b_1 + b_2) \\ z_1 z_2 &= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2). \end{aligned}$$

Fundamentally, in order to be considered addition and multiplication, these definitions should satisfy standard associativity and distribution rules

$$\begin{aligned} (z_1 + z_2) + z_3 &= z_1 + (z_2 + z_3) \\ (z_1 z_2) z_3 &= z_1 (z_2 z_3) \\ z_1 (z_2 + z_3) &= z_1 z_2 + z_1 z_3, \end{aligned}$$

which can easily be shown to be true for complex numbers. In addition, one would like to be able to associate a complex number with a real number known as its magnitude

$$|z_1|^2 = a_1^2 + b_1^2,$$

which amongst other things should satisfy the rule

$$|z_1 z_2| = |z_1||z_2|.$$

In 1843, Hamilton wanted to extend 'couples' (complex numbers) to 'triples' – numbers of the form

$$z = a + ib + jc$$

where $i^2 = -1$ and $j^2 = -1$ (but $i \neq j$). The idea is that if you can make up one 'imaginary' square root of -1, why not make up another in a 'perpendicular' direction. Unfortunately reality turned out to be less enthusiastic about this idea. Apparently, Hamilton's children would ask him at breakfast each morning *"Papa, can you multiply triples yet?"* to which he would reply sadly *"No, I can only add and subtract them"*. In October of that year while out for a walk, the answer finally came to him. He graffitied the answer into the Brougham Bridge. The graffiti has faded over the years, but there is now a plaque commemorating the event if you wish to visit it while in Dublin.

[a] At least, everybody who makes it to a PLANCKS final

(a) [1 mark] Prove that $ij = -ji$, i.e. the different square roots of -1 must anti-commute.

(b) [2 marks] Prove that if one writes $k = ij$, then $k^2 = -1$, i.e. k is another square root of -1. Further prove that k can not be written as a 'triple' i.e. $k \neq \alpha + \beta i + \gamma j$ for real α, β, γ.

You have now shown that 'triples' do not exist if one wants a consistent algebra including multiplication – they must be extended into quaternions.

Hamilton had a notation for quaternions

$$q = (s, \vec{v}) = s + v_x i + v_y j + v_z k,$$

where he called the real part of the quaternion s the 'scalar' part, and the imaginary parts of the quaternion $\vec{v}$ the vector part. In this notation, consider $q_1 = (0, \vec{v}_1)$ and $q_2 = (0, \vec{v}_2)$.

It is worthy of historic note that the modern concept of scalar and vector product had not yet been invented when Hamiltonian first developed his theory of quaternions.

(c) [2 marks] From the rules you have previously derived about multiplying quaternions, prove that

$$q_1 q_2 = (-\vec{v}_1 \cdot \vec{v}_2, \vec{v}_1 \times \vec{v}_2).$$

(d) [5 marks] Calculate e^q if

$$q = \frac{\pi}{5} i + \frac{4\pi}{15} k.$$

Hint: First, work out why this is not trivial. Second, you may find it easier to first work out the general formula for e^q where q is a quaternion, and then substitute in the specific value given.

3 Möbius Strip.... Time Travel?

In Avengers Endgame, Tony Stark asks his AI assistant 'Friday' to find the eigenvalues of an inverted Möbius strip so that they can build a device to time travel into the past and... *spoilers.*

What is a Möbius strip? Imagine a rectangle of length L and width W, you connect the two ends and the result is a band of circumference L. However, if you **twist** the rectangle 180° before you connect the ends, you get a Möbius strip.

In this problem, you will solve the Schrödinger equation for a quantum particle of mass m confined on a Möbius strip.

For this problem, consider the easiest geometry that a space with Möbius strip topology can have- a flat space with Möbius strip boundary conditions. The problem will be further simplified by ignoring spin.

(a) [6 marks] Find the energies and normalised wavefunctions of a quantum particle of mass m moving on a Möbius strip of length L and width W with potential $V(x, y) = 0$ everywhere on the strip.

Now let us imagine that the confined quantum particle is an electron with mass, $m = 9.11 \times 10^{-31}$ kg.

The electron is described at time $t = 0$ by a wave packet with wavefunction confined to a circle of radius a at the point $x = \frac{W}{2}$, $y = 0$ (depending on your orientation of the strip).

Take the length of the Möbius strip, $L = 20$ nm and the width, $W = 3$ nm.

(b) [4 marks] After what time will the wavefunction return to its initial position? Give your answer in seconds.

State any assumptions you have made to get to your answers.

4 Icy Roof

After a night of frivolity, Patrick decides to climb on to the top of his roof. His house has an interesting shape - the upper half of a sphere of radius 10 m. It being an icy night, Patrick unsurprisingly starts to slip down.

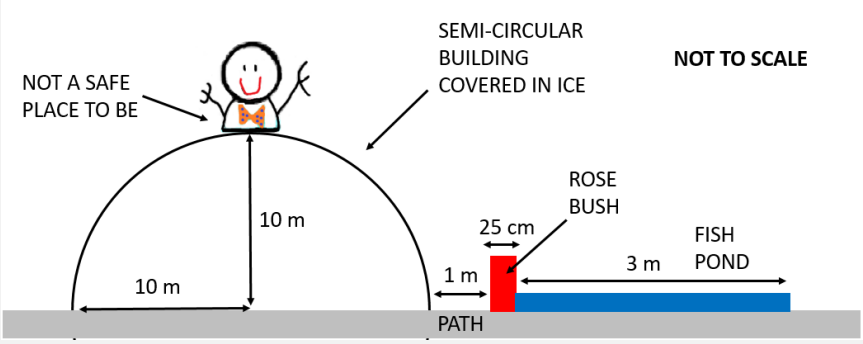


Figure 3: Sketch of Patrick's house and garden.

(a) [6 marks] Making the standard undergraduate physics approximations (no friction, point mass, etc.), calculate how far from the edge of the house Patrick lands, to see if he makes it into the pond, slams into the concrete path, or ends up in the thorny rose bush.

(b) [4 marks] Without detailed calculation (but with sound physics arguments), estimate how much of a difference it makes to relax the more unreasonable approximations to predict where Patrick will really land.

5 Floating Ring

A thin superconducting ring is held above a vertical, cylindrical magnetic rod. The axis of symmetry of the ring is the same as that of the rod. The cylindrically symmetrical magnetic field around the ring can be described approximately in terms of the vertical and radial components of the magnetic field vector as $B_z = B_0(1 - \alpha z)$ and $B_r = B_0 \beta r$, where B_0, α and β are constants, and z and r are the vertical and radial position coordinates, respectively.

A sketch of this can be seen in Figure 4.

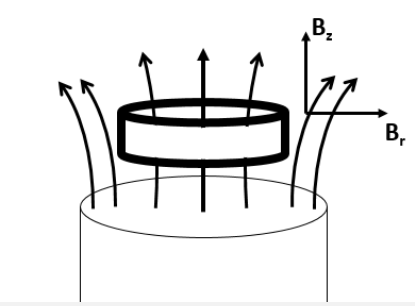


Figure 4: A thin superconducting ring above a cylindrical metal rod.

Initially, the ring has no current flowing in it. The coordinates of the centre of the ring are $(z, r) = (0, 0)$. When released, it starts to move downwards with its axis still vertical.

Useful data:

- Ring's mass, $m = 50$ mg
- Ring's radius, $r_0 = 0.5$ cm
- Ring's inductance, $L = 1.3 \times 10^{-8}$ H
- Magnetic field constant, $B_0 = 0.01$ T
- Magnetic field constant, $\alpha = 2$ m^{-1}
- Magnetic field constant, $\beta = 32$ m^{-1}

(a) [8 marks] Show that the ring undergoes simple harmonic motion, and find the frequency and amplitude of the oscillation.

(b) [2 marks] What is the maximum current that flows in the ring, and where in the oscillation does it occur?

6 Nuclear Medicine Scan

Red deer are native to Ireland, as they get older the male deer grow very large antlers. A young deer wanted to check if his budding antlers were growing uniformly. So, the Stag visited St James' Hospital in Dublin. He received a nuclear medicine scan, whereby he was injected with a photon-emitting radionuclide which accumulates primarily, but not exclusively, in areas of bone growth (his antlers).

The signal from the radioactivity measured from a detector placed outside an object will depend on both the activity present and on the distribution of the photon attenuation coefficients.

Consider an object with a distribution of attenuation coefficients $\mu(x, y)$ and radioactivity distribution $A(x, y)$ as shown in Figure 5.

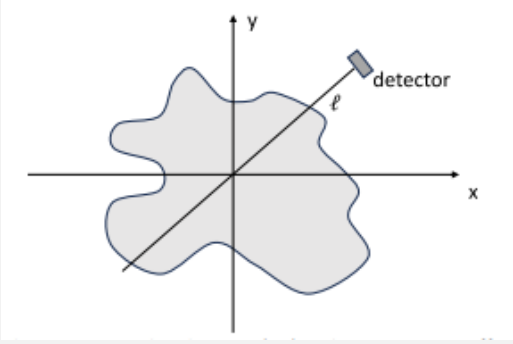


Figure 5: Object with arbitrary activity distribution $A(x, y)$ and attenuation coefficient distribution $\mu(x, y)$.

Assume that the detector has 100% efficiency (i.e., all photons hitting it are detected) and that it is collimated (i.e., it will only detect photons emitted along a thin line). The photons emitted, from a point source, in a specific direction are proportional, by a factor $k < 1$, to the total number of photons emitted.

Remember that the variation in intensity for a beam travelling through an infinitesimally small thickness $\mathrm{d}t$ is

$$\mathrm{d}I = -\mu I \mathrm{d}t.$$

(a) [1 mark] In the absence of attenuation, the signal measured along an arbitrary line ℓ is given by

$$I = k \int_0^{\ell} A(x, y)\, \mathrm{d}s.$$

How would this be modified when attenuation is taken into account? Assume that there is no geometrical correction for the distance of the emission point from the detector.

Consider now the top of the Stag's skull, approximated by an ellipse with semi-axes a and b, arbitrary radioactivity distribution $A(R, \vartheta)$ in polar coordinates and a constant attenuation coefficient μ (see Figure 6).

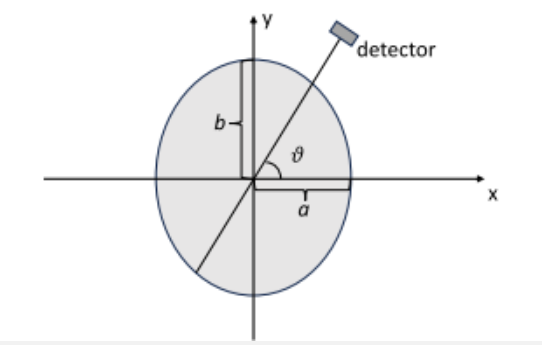


Figure 6: Ellipse with arbitrary activity distribution and constant attenuation coefficient.

(b) [3 marks] Write an expression, in polar coordinates, for the signal measured from a line going through the origin of the axes at an angle ϑ to the x-axis.

So far we have assumed a point-like detector with infinite angular resolution (i.e., it will only detect photons emitted along a very thin line). The detector now has a circular aperture of radius R, as shown in Figure 7).

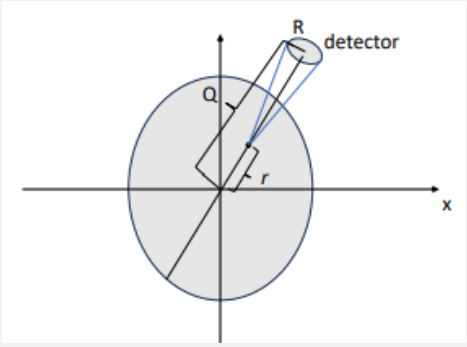


Figure 7: Activity distribution in a plane, with a circular detector aperture

We can assume that the radioactivity is still distributed in a plane, but that photons are emitted isotropically in a sphere.

(c) [3 marks] How would your solution to part b change if we had a detector with a circular aperture of radius R?

The diagram in Figure 8 shows the Stag's scan, with a diffuse "background" uptake of radionuclide and two "points" representing the antler buds.

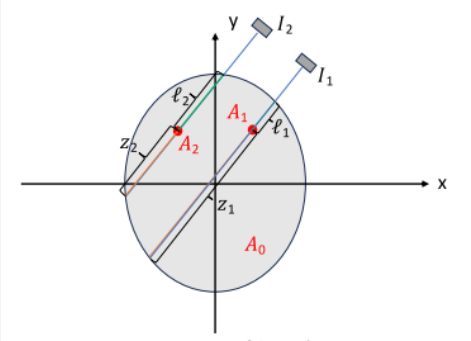


Figure 8: Diagram of the Stag's scan.

- The detector counts measured along the two parallel lines are $I_1 = 1.23$ s^{-1} and $I_2 = 1.15$ s^{-1}.
- The two antler buds can be treated as point-like, with radioactivity uptakes A_1 and A_2. If $A_1 = A_2$, the antlers were growing uniformly and the Stag will now have developed a pair of symmetrical antlers.
- The radioactivity/unit length in the rest of the skull is $A_0 = 3000$ Bqcm^{-1}.
- The attenuation coefficient of the head tissue is 0.27 cm^{-1}.
- $\ell_1 = 5.25$ cm, $\ell_2 = 11$ cm, $z_1 = 24.5$ cm and $z_2 = 12.5$ cm
- For this part of the question, assume again a perfectly collimated detector, with the ratio between photons emitted in the detector's direction/total photons emitted at each point $k = 10^{-4}$.

(d) [3 marks] On the basis of the quantities above, calculate A_1 and A_2 and determine if the Stag's antlers were growing uniformly at the time of the scan.

7 Dark Matter in the Galaxy

In a galaxy all objects inside orbit around its centre. At a point $r = 2.6 \times 10^4$ ly from the centre the measured velocity of orbit is $v_{meas} = 250$ kms^{-1}. However, if we were to calculate the velocity of orbit at distance r based on all the luminous matter in the galaxy, the value would be much lower, $v_{calc} = 15$ kms^{-1}. Why is there such a large difference? It can only be possible if the galaxy had a lot more matter inside but it was hidden from view. The hidden matter is now called dark matter and we know it exists throughout the universe, even though we have no idea what it is.

Imagine you are an astronomer in this galaxy who is investigating dark matter in the galaxy's spherical halo.

Useful data:

- Gravitational constant, $G = 6.67 \times 10^{-11}$ Nm^{2}kg^{-2}
- One light year, 1 ly $= 9.46 \times 10^{15}$ m
- One electron volt, 1 eV $= 1.602 \times 10^{-19}$ J

(a) [5 marks] Calculate the average dark matter density in this region of the galaxy. Give your answer in $\frac{\text{GeV}}{\text{c}^2}$cm^{-3}.

Your galaxy is in a universe where where all the dark matter is made up of miniature black holes. These mini black holes are spread throughout a galaxy's spherical halo (like a dark matter halo).

The mini black holes orbit around the centre of the galaxy like every other object. Your planet sits approximately at distance r from the centre of the galaxy. Through observation you notice that roughly once a year one of these mini black holes pass between you and your local star which you can assume is at distance ~ 1 AU to you.

Useful data:

- One astronomical unit, 1 AU $= 1.50 \times 10^{11}$ m
- Solar mass, 1 M$_\odot$ $= 2 \times 10^{30}$ kg

(b) [5 marks] What is the approximate mass of these miniature black holes? Give your answer in units of the Solar mass.

State any assumptions you have made to get to your answers.

8 Icosahedron of Resistors

An icosahedron consists of 20 equilateral triangles. It has 12 vertices and 30 edges, with 5 edges meeting at each vertex. Figure 9 shows an icosahedron and its net.

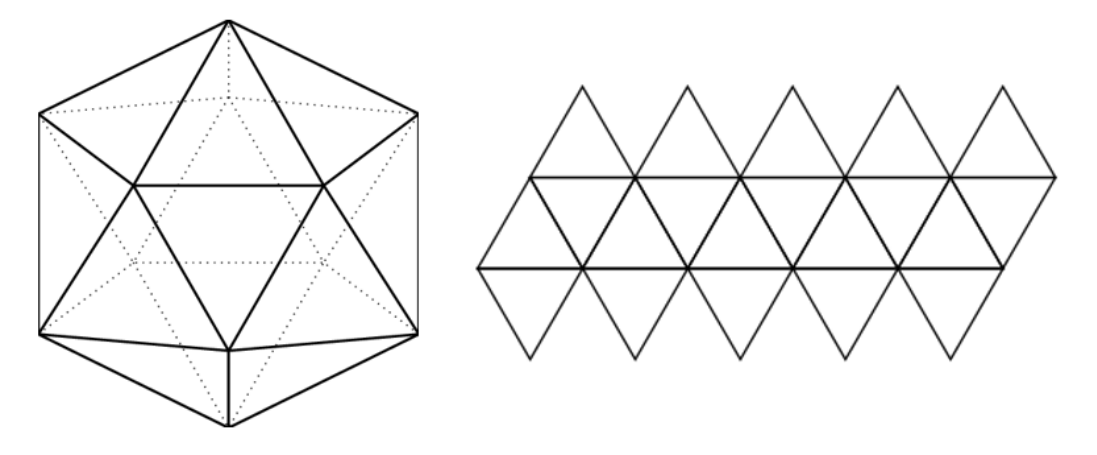


Figure 9: Icosahedron (left) and its 2D net (right).

Now imagine that this icosahedron was a component in a circuit, where each edge of the icosahedron is a 1Ω resistor.

You may use the *model icosahedron kit* provided to you.

(a) [10 marks] Find the effective resistance between two adjacent vertices.

9 Molecular Zipper

In the 1960s, Charles Kittel proposed a toy model to illustrate the physics of the separation of strands of DNA. The model consists of two long molecules coupled by N links, with the following rules:

- Each link can be closed or open – and if open, they can be in one of G orientations where $G > 0$ is an integer parameter of the model. There is only one way the link can be closed.
- If links 1 to n are open, then link $n + 1$ can also open with energy cost $\epsilon > 0$. This is the basic idea of the zipper model – a link can only be open if all the ones before it are also open.
- The final link $n = N$ can not be opened.

The final rule means that the model is a 'single-ended zipper' – which is a slight simplification of earlier models that could be 'unzipped' from both ends.

(a) [4 marks] Calculate the free energy of the model with N links at a temperature T.

(b) [2 marks] Hence show that the model has a finite-temperature phase transition if $G > 1$, and find an expression for the transition temperature.

(c) [2 marks] Calculate the expected number of open links as a function of temperature, and show that this may be used as an order parameter for the phase transition.

(d) [2 marks] Come up with a physical explanation why one requires this extra degeneracy $G > 1$ in order to have a phase transition in the model.

10 4D Sun

Imagine you were transported to another universe with one more spatial dimension, everything appeared as 4D. In this problem, you will investigate if the light generated from the Sun would be different in this universe.

You may assume that the surface temperature of the Sun would not change from our universe and is approximately $T = 6 \times 10^3$ K.

Useful data:

- Boltzmann constant, $k_B = 1.38 \times 10^{-23}$ JK^{-1}
- Speed of light, $c = 3 \times 10^8$ ms^{-1}

(a) [10 marks] What would the colour of the Sun be in this four-dimensional universe?









Thank you to the following people for their help and contributions to this exam:

Alexia Beale – University of Surrey & PLANCKS Advisory Board

Dr Sam Carr – University of Kent & PLANCKS Advisory Board

Professor Emeritus Alan Davies - University of Hertfordshire

Emma Elley - University of Oxford & PLANCKS Advisory Board

Dr James Kneller – Northeastern University London & PLANCKS Advisory Board

Dr David Morin - Harvard University

Prof Alexander Morozov - University of Edinburgh

Dr Silvia Pani – University of Surrey

Anthony Quinlan – PLANCKS Advisory Board

Jack Woodward – University of Liverpool & PLANCKS Advisory Board

A special thanks to the following people whose work served as reference and inspiration:

Prof A. De Angelis, Dr M. Pimenta, Dr R. Conceição, Dr P. Gnädig, Dr G. Honyek & Dr K. F. Riley